

RETAIL BUSINESS ANALYSIS

Data Analysis Project Report

1. Introduction

This project analyzes retail transaction data to understand sales performance, profitability, product performance, regional performance, discount patterns, and monthly sales trends. Python, SQL, and Tableau were combined to turn transaction data into practical business insights.

2. Abstract

The Superstore dataset was analyzed to identify factors associated with retail sales and profit. Exploratory analysis was performed in Python, SQL was used for business-level aggregation, and an interactive Tableau dashboard was created. Technology was the strongest category by profit, West was the strongest region, and higher discount levels were associated with weaker or negative profit. Profitable and loss-making sub-categories and seasonal sales patterns were also identified.

3. Tools Used

Python: Pandas, NumPy, Matplotlib, Seaborn for cleaning, calculations and visualizations.

SQL: SQLite for aggregation and profitability queries.

Tableau Public: Interactive dashboard and visual analysis.

4. Steps Involved in Building the Project

1	Dataset preparation	Loaded the Superstore dataset with 9,994 records and 21 columns.
2	Data inspection	Checked structure, types and missing values; no missing values were found.
3	Python EDA	Analyzed profit margin, categories, sub-categories, regions, discounts, products and monthly sales.
4	SQL analysis	Created queries for sales, profit, margin, region, discount and top/bottom products.
5	Tableau	Built and published visuals for sales/profit, discount, category, products, sub-categories, monthly trends and regions.
6	Insights	Converted analytical results into business recommendations.

5. Key Findings

- Total sales were approximately **2.30 million** and total profit approximately **286 thousand**.
- **Technology** had the highest category profit (145,455) and profit margin (17.40%).
- **Furniture** had the lowest category profit margin (2.49%) despite substantial sales.
- **West** had the highest regional profit (108,418) and margin (14.94%); **Central** had the lowest margin (7.92%).
- Discounts of **30% or more** were associated with negative profit at several discount levels, so high-discount transactions should be reviewed carefully.
- November showed the strongest overall monthly sales pattern, while February was among the weakest months.
- Loss-making sub-categories included **Tables, Bookcases and Supplies**; strong contributors included **Copiers, Accessories, Paper and Phones**.

6. Business Recommendations

- Prioritize profitable Technology products while protecting margins.
- Review pricing, costs and discount policies for Furniture and loss-making sub-categories.
- Monitor high-discount transactions and use discounts selectively.
- Investigate Central-region performance because it has meaningful sales but the lowest margin.
- Prepare inventory and marketing plans ahead of strong November–December demand.
- Review persistent loss-making products for pricing, supplier-cost reduction, rationalization or reduced promotion.

7. Dashboard and Deliverables

The Tableau Public dashboard presents total sales and profit, discount versus profit, category sales and profit, top products, sub-category profitability, monthly sales trend, and regional sales and profit. The project also includes a SQL query file containing the main business-analysis queries.

8. Conclusion

The project demonstrates how Python, SQL and Tableau can be combined for retail business analysis. The results identify profitable categories and products, regional differences, discount-related profitability patterns and seasonal sales behavior, supporting better pricing, product and regional decisions.

Note: The project guide mentions inventory-days analysis, but the selected Superstore dataset does not contain an Inventory Days field. No inventory-days values were invented; the analysis uses the fields available in the dataset.